FTC Merger Enforcement and HSR Transaction Volume Analysis paper(FY 1996 - 2020)

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Research Question, Data, and Methods

This project examines how FTC merger enforcement activity changed over time relative to HSR transaction volume and how enforcement varied across industries. It also explores whether the industry composition and types of enforcement actions differed between earlier and later periods.

The analysis uses two datasets published by the Federal Trade Commission.[1] The first contains annual HSR transaction and filing statistics. I use the variable *TransactionsReported* to measure annual merger transaction volume. Although the original plan proposed aggregating monthly data, I selected the annual dataset to ensure a consistent measure of transactions. The second dataset contains 433 public FTC merger enforcement records, including the fiscal year, action date, matter number, case name, industry, case link, and enforcement type. The final analysis covers FY1996–2020, with 425 enforcement records.

I conducted exploratory data analysis in Python using pandas, matplotlib, and seaborn. Data preparation included checking missing values and duplicate records, labeling missing industries as "Unknown," and grouping industry subcategories into broader sectors. I merged annual enforcement counts with HSR transaction counts by fiscal year. The analysis compares annual trends, year-over-year percentage changes, enforcement actions per 1,000 transactions, and correlation. Industry and enforcement type shares are compared between FY1996–2007 and FY2008–2020, using percentages to account for the periods' different lengths.

Main Findings

Trends in Transaction Volume and Enforcement

Annual HSR transaction volume and FTC merger enforcement activity did not consistently move together. From FY2008 to FY2009, reported transactions fell from 1,726 to 716, a decrease of 58.5%. During the same period, FTC enforcement actions increased slightly from 15 to 16, or 6.7%. The year-over-year comparison shows that changes in merger activity were not always accompanied by changes in enforcement in the same direction or of a similar magnitude.

Enforcement actions per 1,000 HSR transactions also fluctuated. The ratio reached approximately 22.35 in FY2009, compared with 8.69 in FY2008. This increase largely reflected the sharp decline in transaction volume while enforcement counts remained relatively stable.

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[1] Federal Trade Commission, "Data Sets," annual HSR transaction data and FTC merger enforcement action data, https://www.ftc.gov/policy-notices/open-government/data-sets.

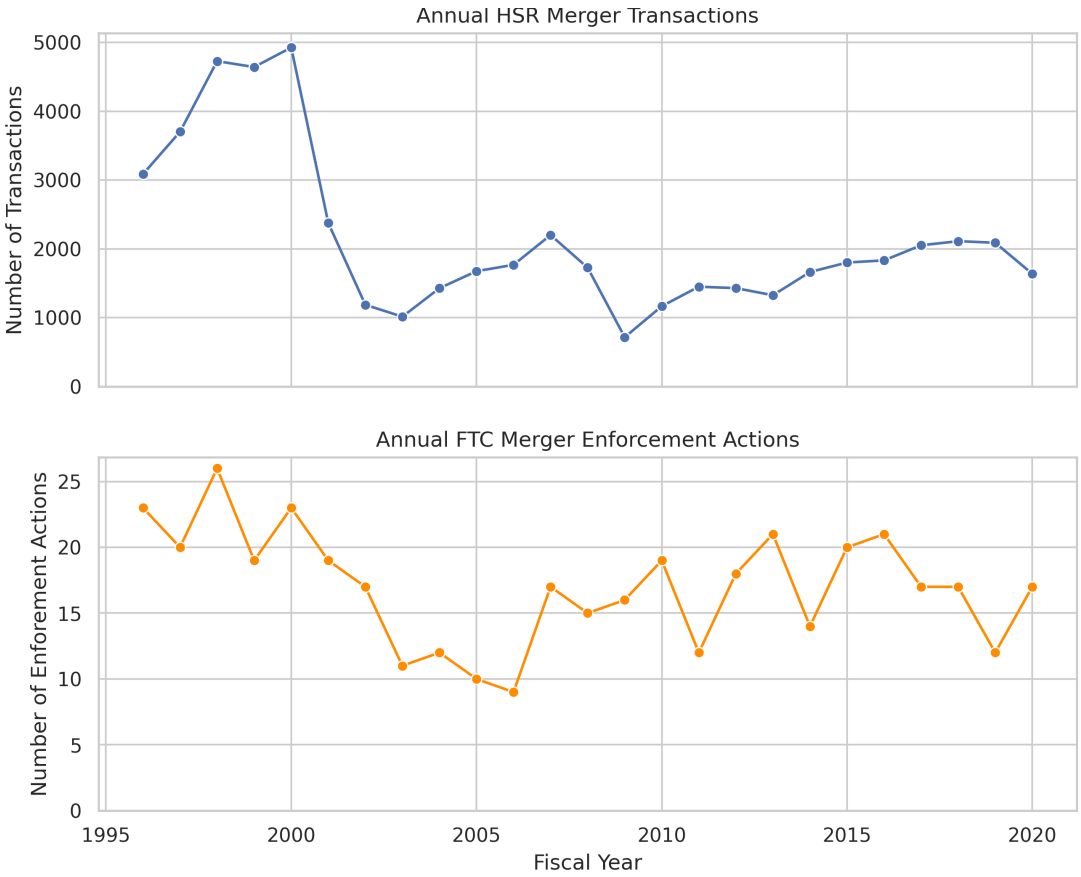


Figure 1. Annual HSR transactions and FTC merger enforcement actions(FY 1996-2020) [Source : Author's analysis of FTC dataset]

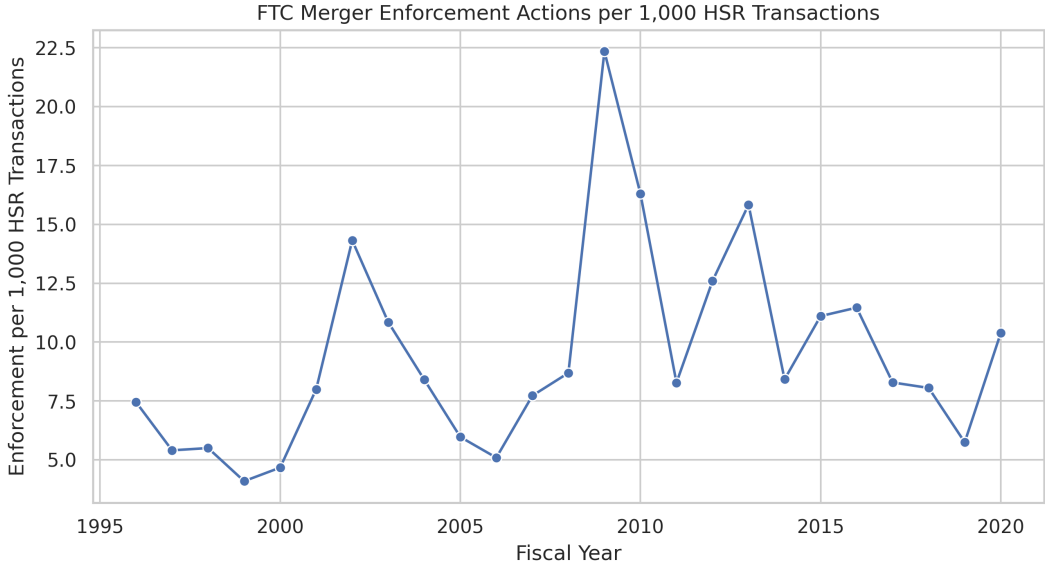


Figure 2. FTC merger enforcement actions per 1,000 HSR Transaction (FY 1996-2020) [Source : Author's analysis of FTC dataset]

Therefore, an increase in the ratio alone does not demonstrate more aggressive enforcement.

The Pearson correlation[2] between annual transaction volume and enforcement counts was 0.59, indicating a moderate positive linear association. Years with higher transaction volumes generally had more enforcement actions. However, this overall association coexisted with substantial differences in annual changes and does not establish causation.

Differences Across Industries

Health care accounted for the largest share of enforcement actions across FY1996–2020: 153 of 425 records, or 36.0%. Manufacturing followed with 118 actions, or 27.8%, and energy with 52 actions, or 12.2%. Together, these sectors accounted for approximately 76% of all enforcement records.

The industry composition changed between the two periods. Health care's share increased from 25.7% in FY1996–2007 to 45.7% in FY2008–2020, with counts rising from 53 to 100. Energy's share declined from 17.0% to 7.8%, professional services from 7.3% to 3.2%, and defense from 5.8% to 0.5%. Manufacturing's share decreased modestly, while retail and information and technology remained relatively similar. These comparisons describe the composition of enforcement, rather than industry-specific enforcement rates.

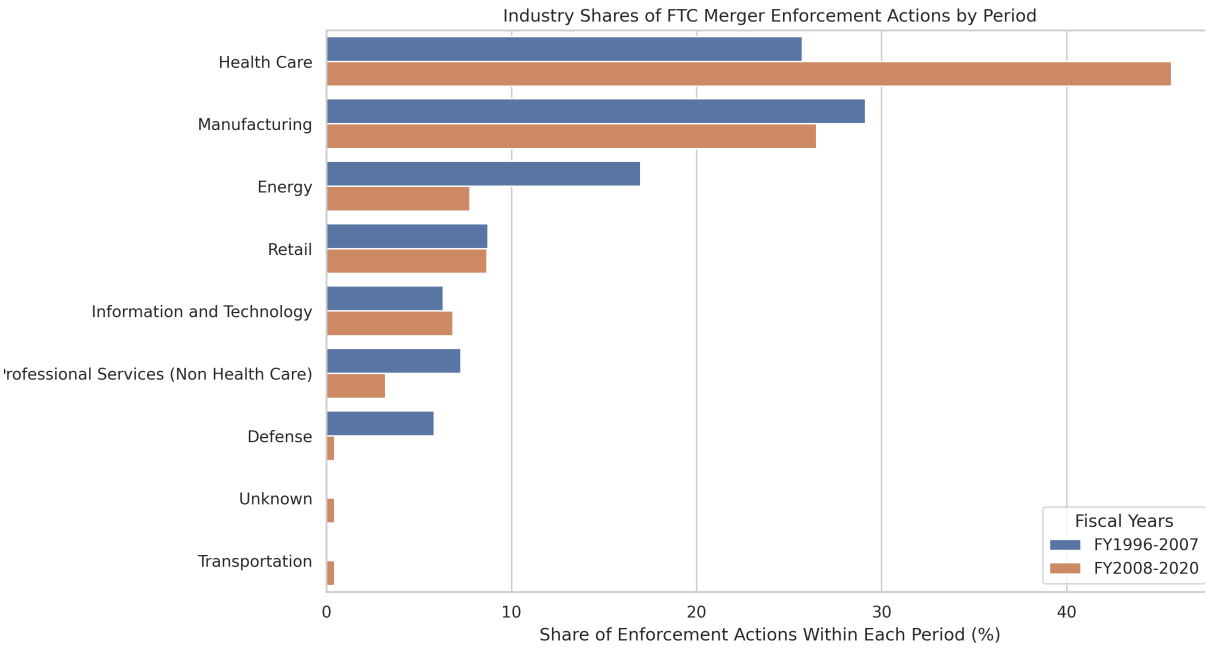


Figure 3. Industry shares of FTC merger enforcement actions within each period

[Source : Author's analysis of FTC dataset]

[2] pandas, "pandas.Series.corr," https://pandas.pydata.org/docs/reference/api/pandas.Series.corr.html.

Changes in Enforcement Types

“Consent Order Accepted for Comment” remained the dominant enforcement category in both periods. Its share declined from 84.0% to 79.0%, although its count was unchanged at 173 actions in each period. Preliminary injunctions increased from 26 to 37 actions, with their share rising from 12.6% to 16.9%. Administrative complaints and permanent injunctions represented much smaller shares. These results indicate a modest change in the mix of recorded enforcement actions.

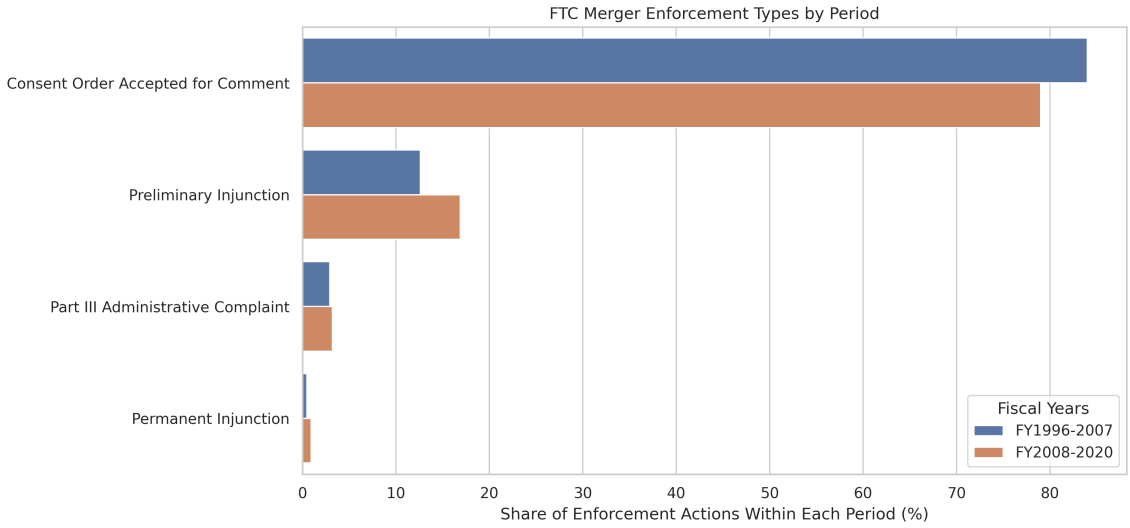


Figure 4. Shares of FTC merger enforcement types within each period

[Source : Author’s analysis of FTC dataset]

Limitations and Policy Implications

This analysis has several limitations. First, the HSR transaction dataset does not provide industry-level transaction counts. Consequently, industry comparisons describe shares of FTC enforcement actions rather than enforcement rates relative to merger activity within each sector. Second, the enforcement dataset does not provide the full timeline from transaction filing or investigation opening to resolution. It therefore cannot show how long individual matters took or explain delays between transaction reporting and enforcement. The annual ratio is also descriptive: FTC enforcement actions and HSR transactions do not necessarily represent the same transactions in the same year. HSR transaction totals cover transactions reported to both federal antitrust agencies, whereas the enforcement records cover only the FTC. In addition, the broad industry categories may conceal differences between subindustries, and the period comparisons cannot identify when changes occurred or whether they were sustained throughout each period.

The findings suggest that enforcement monitoring should consider transaction volume, industry composition, and enforcement types together. Health care's growing share of enforcement warrants further examination, but does not by itself establish that mergers in that sector became more problematic.

The increased share of preliminary injunctions also raises questions about the role of timely intervention. In rapidly changing technology markets, including artificial intelligence, further research could examine when preliminary injunctions are appropriate and whether they effectively preserve competition. This study does not establish that greater use is necessary, but identifies an issue for future policy analysis.